Giuseppe Baldassarre

Geospatial Analyst | Remote Sensing & Earth Observation

Ph.D. Environmental Engineering (Remote Sensing) · M.Sc. Mediterranean Forestry and Natural Resource Management

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Portfolio: gbal17.github.io · **Full online CV: gbal17.github.io/taleo-cv.html**

LinkedIn · Google Scholar · ResearchGate · Medium · GitHub

Remote Sensing and GIS Specialist with over twelve years of professional experience, I design and operationalize satellite-based monitoring systems for land, water and agriculture. I combine advanced GIS/RS processing (Python, Google Earth Engine, SEPAL) with decision-support tools and inclusive governance mechanisms to support FAO, IFAD-funded and UNCCD projects and national partners in climate-vulnerable countries.

Skills

Languages: Italian (mother tongue), English (excellent), Spanish (excellent working knowledge), Portuguese and Turkish (intermediate)

Programming & data: Python (GeoPandas, Rasterio, scikit-learn, XGBoost, pandas), Google Earth Engine (JavaScript & Python API, geemap), MATLAB, Git/GitHub, LaTeX

Platforms & tools: QGIS, PyQGIS, SEPAL, Collect Earth Online, EarthMap, FAO Hand-in-Hand Geospatial Platform, Office

Domains: Optical and SAR remote sensing, land cover classification (LCML/ISO 19144-2), change detection, time-series analysis, land degradation (SDG 15.3.1), accuracy assessment, spatial data harmonization and metadata

Professional experience

Geospatial Analyst — *FAO, Land and Water Division* — 2023 – present

- Designed the geospatial indicator framework for the **IFAD-funded ROOTS project** (Resilience of Organizations for Transformative Smallholder Agriculture, UTF/GAM/047/GAM), The Gambia (read); produced the 2023 Land Cover Map of The Gambia, the first under the LCML/ISO 19144-2 reference system (read, watch), and built the national geospatial monitoring platform now used by government counterparts
- Leading the GEE-to-PRAIS4 automation pipeline for SDG 15.3.1 / Land Degradation Neutrality reporting across **21 UNCCD country Parties** (2026 PRAIS4 cycle, COP17 deadline); piloted the full workflow for Mongolia (land cover, LPD, SOC, transition matrices)
- Designed the district-level INFORM Risk Index for FAO Afghanistan's AWARE platform (∼401 districts, 60+ indicators); published 35 composite layers to the CKAN catalogue
- Produced Syria's 10 m national land cover map (AlphaEarth Foundations embeddings + Sentinel-2 + Random Forest, GEE) for drought impact assessment (read)
- Flood impact assessment on cropland in Pakistan, Oct–Nov 2025 (KP, Punjab, Sindh)
- Land Cover Atlas of Yemen (read)
- Development of the Gambia GeoLab geospatial platform, presented at the Geo for Good Summit in Singapore, 2025 (FAO news, read)
- Design and delivery of trainings on "Geospatial Tools for Mangrove Monitoring in The Gambia" (visit) and "Using SEPAL, Google Earth Engine and Collect Earth Online to produce a Land Cover Map of Gambia (2023)" (visit)

Post-doctoral researcher — *University of Lisbon, Portugal* — 2020 – 2023

- Estimating shrub live fuel moisture content for mainland Portugal (read)
- Study of traditional extensive pastoralism relying on fires in Northern Portugal with GIS/RS analysis
- Mapping fire regimes over Portugal (read1, read2) and Turkey (read)
- Automatic monitoring of the national fuel break network of Portugal with Sentinel-2 in GEE (read)

EU Aid Volunteer — *Sustainable Rural Development, Viet Nam* — 2021 – 2022

- SAR-based above-ground biomass and carbon stock estimation for mangrove restoration (VM069), with a GEE application to monitor mangrove extent in Southern Viet Nam
- Developing an App in GEE to monitor tree canopy cover in Viet Nam (2000–2021)
- Mapping smallholder coffee production systems driving land use change in NW Viet Nam (watch)

Post-doctoral researcher — *Istanbul Technical University, Turkey* — 2012 – 2018

- Biomass burning emission estimation with geostationary observations (read)
- Description of forest fires over Turkey using the most advanced satellite observations (visit)
- First version of the national Forest Fire Monitoring System web-GIS in Turkey (UYANBIL) (visit)

Environmental consultant (volunteer) *Aprode Peru (NGO), Peru* 2011

- Field campaigns to outline the solid waste management cycle in rural villages in Northern Peru
- Implemented a project for the social inclusion of informal recyclers (watch)

Visiting scholar *University of Wisconsin–Madison, USA* 2008, 2009 – 2010

- Development and assessment of advanced satellite fire detection products using statistical tools
- Testing product operational applicability over Southern Italy
- Publications: "Automatic RST-based system for a rapid detection of fires"; "Assessment of the Robust Satellite Technique (RST) in real-time detection of summer fires"

Education

M.Sc., Mediterranean Forestry and Natural Resource Management *Erasmus Mundus MEDfOR* 2018 – 2020

- 1st year: University of Lisbon (Portugal), Valladolid (Spain) & Padova (Italy) — natural resource management with decision support systems, forest governance, agroforestry, fire ecology & management, GIS, research & project development methodology
- 2nd year: University of Porto (Portugal) & Lleida (Spain) — governance of Mediterranean forests, environmental economics, research methods
- Thesis: "Describing fire regimes over Turkey using MODIS fire observations". Graduated with excellent standing (8.91/10)

Ph.D., Environmental Engineering (Remote Sensing) *U. of Wisconsin–Madison & U. of Basilicata* 2007 – 2011

- Advanced tools for satellite-based environmental monitoring (biomass combustion emissions)
- Thesis: "Performance of the most advanced satellite techniques for real-time monitoring of forest fires over Italy"
- Publication: "A Total Validation Approach for assessing the RST technique in forest fire detection and monitoring"

M.Sc., Environmental Engineering (110/110 *summa cum laude*) *University of Basilicata, Italy* 2004 – 2007

- Thesis: "Robust satellite techniques for fire monitoring using geostationary satellites (MSG-SEVIRI)"
- Waste-water and waste management; atmospheric physics; water, air and soil pollution dynamics
- Publication: "Assessment of a Robust Satellite Technique for forest fire detection and monitoring using a Total Validation Approach (TVA)"

B.Sc., Environmental Engineering (110/110 *summa cum laude*) *University of Basilicata, Italy* 2001 – 2004

- Physics, chemistry, geology, statistics, economics, applied environmental engineering

Supporting documents: certificates, grades, extra-courses, international cooperation, references.